Done

- [ ] Basics sharpening
- [ ] Non abelian papers true gaps that need to be addressed
- [ ] Tensor paper theory
- [ ] What are needed to address all of the gaps

Basics Sharpening:

PART 7 — Representation

A representation means:

“Represent symmetry operations using matrices.”

Instead of abstract rotations,

we use matrices acting on vectors/states.

Example:

rotation by angle θ:

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

This is a representation.

PART 8 — Reducible vs Irreducible

Suppose you have:

$$(s, p_x, p_y)$$

Under rotation:

- s stays alone
- p_x, p_y mix together

So the matrix looks like:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & * & * \\ 0 & * & * \end{pmatrix}$$

Notice:

the s part separates.

So this representation can be split.

This is called:

reducible representation

But the (p_x, p_y) pair cannot be split further.

They always mix together.

That smallest indivisible block is:

irreducible representation (irrep)

2. What Is an Irrep REALLY?

Irrep means:

“Smallest set of things that always mix together under symmetry.”

That’s it.

Example 1 — Single object

Suppose:

$$s \rightarrow s$$

under every symmetry operation.

It never mixes with anything.

So it forms a 1-dimensional irrep.

Why 1D?

Because only ONE object exists.

3. What Does “Dimension” Mean?

Dimension of irrep means:

“How many components mix together.”

1D irrep

One object alone.

Example:

$$s$$

Under symmetry:

$$s \rightarrow (+1)s$$

or maybe

$$s \to (-1)s$$

Still only ONE object.

So:

- representation matrix is 1×1
- irrep dimension = 1

2D irrep

Two objects mix together.

Example:

$$(p_x, p_y)$$

Under rotation:

$$p_x \to ap_x + bp_y$$

$$p_y \to cp_x + dp_y$$

Now we need:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

a 2×2 matrix.

So this is a 2D irrep.

4. MOST IMPORTANT INTUITION

1D irrep

Things only gain a PHASE/sign.

Example:

$$f \to +f$$

or

$$f \to -f$$

No mixing.

2D irrep

Objects ROTATE into each other.

Example:

$$(x, y)$$

or

$$(p_x, p_y)$$

or

$$(E_x, E_y)$$

9. What Happens in Abelian Groups?

VERY IMPORTANT:

In abelian groups:

all irreps are 1D

This is huge.

Why?

Because all symmetry operations commute.

So nothing truly “mixes.”

Every object can be labeled independently.

Only phases/signs appear.

11. Example — Reflection Group

Group:

$$\{+1, -1\}$$

Representations:

Symmetric irrep

$$f \to +f$$

Antisymmetric irrep

$$f \to -f$$

Again:
1D irreps only.

14. Why Non-Abelian Creates Higher-Dimensional Irrep

Because operations do not commute,

objects genuinely mix.

Example:

$$(p_x, p_y)$$

A rotation continuously rotates them into each other.

You CANNOT separate them.

So they form a 2D irrep.

16. Example — SU(2)

Spin symmetry.

Non-abelian.

Generators:

$$[S_x, S_y] = iS_z$$

Notice:

$$S_x S_y \neq S_y S_x$$

So non-abelian.

Then irreps are:

- spin-0 → 1D
- spin-1/2 → 2D
- spin-1 → 3D
- spin-3/2 → 4D

5. What Does "Reduce to Abelian" Mean?

Take a non-abelian irrep:

$$E = (E_x, E_y)$$

Under full symmetry:

- components mix

But in an abelian subgroup,
mixing disappears.

Then:

$$E \to B_1 \oplus B_2$$

Now each component treated independently.

So:

$$E_x \to B_1$$

$$E_y \to B_2$$

Now everything becomes 1D irreps.

Easy for computers.

7. What Happens Physically?

The TRUE symmetry object was:

$$(E_x, E_y)$$

a coupled pair.

But abelianization artificially says:

$$E_x,\ E_y$$

are independent.

This is NOT fully correct physically.

It loses coupling structure.

Non abelian papers true gaps that need to be addressed

The paper's own gaps (proves the problem, doesn't solve it)

Gap 1 — operator pool is incomplete. SymUCCSD discards cross-component excitations, the remaining operators all commute, and the reachable set collapses to a torus instead of the full unitary group. Fix in principle: put the off-diagonal operators back.

Gap 2 — gradient plateau. Even with a complete pool, standard Abelian-adapted HF gives zero cross-component integrals, so the optimizer sees zero gradient along the restored directions and never moves. Fix in principle: break the basis symmetry.

The paper's own key point: these two are independent. Fixing one without the other still fails.

The deeper open gaps

Gap 3 — the symmetry-breaking rotation is not prescribed. The paper says "rotate the orbitals" but gives no derivation of which rotation to apply. A principled answer requires the full non-Abelian Clebsch-Gordan structure to determine the rotation direction canonically rather than by guessing.

Gap 4 — straddling the Fermi level. The easy rotation works only when a degenerate shell sits entirely on one side — all occupied or all virtual. When components are split across the Fermi level (open-shell radicals, transition-metal d-manifolds, O_2's π^* shell), any rotation mixes occupied and virtual orbitals, corrupting the reference. The paper says nothing about this case.

Gap 5 — beyond 2-dimensional irreps. NH_3's E irrep is 2-dimensional and fixed by one angle. The paper states the deficit formula holds for any group but never works out the 3-dimensional cases — Td (methane), Oh (octahedral complexes), Ih (fullerenes) — where far more operators are lost and the restoration requires rotations across a multi-component manifold.

Gap 6 — parameter efficiency. The brute-force fix throws symmetry adaptation away entirely and returns to full UCCSD parameter counts. Recovering completeness cheaply inside an adaptive scheme like ADAPT-VQE, without the algorithm re-trapping itself on the torus by selecting only Abelian-adapted operators, is the hardest and highest-impact open problem.

Tensor paper theory

The Simplest Example: Rotations of an Equilateral Triangle — C3v

Take NH3. The nitrogen sits at the top, three hydrogens form the base. The molecule looks the same after:

- E — do nothing
- C_3 — rotate 120° around the vertical axis
- C_3^2 — rotate 240°
- $\sigma_v, \sigma_v', \sigma_v''$ — reflect through three vertical planes

These six operations form the group C3v. This is your group for NH3.

When you combine two operations you get another one in the list. For example rotating 120° then reflecting gives you another reflection. The group is closed.

PART 2: Representations — How Groups Act On Vectors

The abstract group is just a list of operations. A representation makes it concrete.

A representation assigns a matrix $D(g)$ to every group element g, such that if you combine operations g_1 and g_2:

$$D(g_1 g_2) = D(g_1)D(g_2)$$

The matrices multiply the same way the operations combine. This is the key requirement.

Why do we care? Because quantum states are vectors, and symmetry operations act on them as matrices. The representation tells you exactly how.

Example: C3v Representations

C3v has three irreducible representations (irreps) — the smallest building blocks that cannot be broken down further.

A_1 irrep (dimension 1): Every group element maps to the number 1. Nothing changes.

$$D^{A_1}(E) = 1, \quad D^{A_1}(C_3) = 1, \quad D^{A_1}(\sigma_v) = 1$$

The totally symmetric irrep. The ground state of NH3 transforms as A_1.

A_2 irrep (dimension 1): Rotations give 1, reflections give -1.

$$D^{A_2}(E) = 1, \quad D^{A_2}(C_3) = 1, \quad D^{A_2}(\sigma_v) = -1$$

E **irrep (dimension 2):** Every group element maps to a 2×2 matrix. For example:

$$D^E(C_3) = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}, \quad D^E(\sigma_v) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

This is a genuine rotation and reflection acting on a 2D space. The two degenerate orbitals ϕ_x and ϕ_y of NH3 live here — the symmetry mixes them.

Dimension of an irrep = how many states live in it. A_1 and A_2 have one state each. E has two states. These two E states are always degenerate in energy — symmetry forces it.

PART 3: What SU(2) Is — Simply

SU(2) is the symmetry group of a spinning particle.

When you rotate a spin-1/2 particle (like an electron) in 3D space, the quantum state transforms. The set of all those transformations forms the group SU(2). It is the symmetry group of angular momentum.

The generators of SU(2) are the three spin operators $\hat{S}_x, \hat{S}_y, \hat{S}_z$. They satisfy:

$$[\hat{S}_x, \hat{S}_y] = i\hat{S}_z, \quad [\hat{S}_y, \hat{S}_z] = i\hat{S}_x, \quad [\hat{S}_z, \hat{S}_x] = i\hat{S}_y$$

These commutation relations completely define the algebra. Everything about SU(2) — all its irreps, all its CG coefficients — follows from these three equations alone.

The irreps of SU(2) are labeled by $S = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \ldots$

Each irrep S has dimension $2S+1$. The states inside it are labeled by $m_S = -S, -S+1, \ldots, +S$.

Irrep S	Dimension	States	Physical meaning
0	1	$	0,0\rangle$
1/2	2	$	\uparrow\rangle,
1	3	$	1,+1\rangle,
3/2	4	four states	Spin-3/2

Think of an **irrep** (irreducible representation) of SU(2) as a “complete spin system.”

In quantum mechanics, SU(2) describes **spin rotations**.

1. What does S mean?

S is the **total spin quantum number**.

Possible values are:

$$S = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \dots$$

Examples:

- $S = 0$ → spin-0 particle
- $S = \frac{1}{2}$ → electron-like spin
- $S = 1$ → vector spin
- $S = \frac{3}{2}$ → higher spin system

2. Why dimension $2S + 1$?

For each total spin S, there are multiple orientations of the spin along the z-axis.

Those orientations are labeled by:

$$m_S = -S, -S + 1, \dots, +S$$

The number of allowed values is:

$$2S + 1$$

So the irrep dimension is $2S + 1$.

3. Example: Spin-$\frac{1}{2}$

For

$$S = \frac{1}{2}$$

the allowed m_S values are:

$$m_S = -\frac{1}{2}, \quad +\frac{1}{2}$$

So there are **2 states.**

$$2S + 1 = 2\left(\frac{1}{2}\right) + 1 = 2$$

States are usually written as:

$$\left|\frac{1}{2}, +\frac{1}{2}\right\rangle \quad \text{and} \quad \left|\frac{1}{2}, -\frac{1}{2}\right\rangle$$

Physicists shorten them to:

$$|\uparrow\rangle, \quad |\downarrow\rangle$$

These mean:

- spin up
- spin down

So a spin-$\frac{1}{2}$ system is a 2-dimensional vector space.

4. Example: Spin-1

Now let

$$S = 1$$

Then:

$$m_S = -1, 0, +1$$

There are 3 states:

$$|1, -1\rangle, \quad |1, 0\rangle, \quad |1, +1\rangle$$

Dimension:

$$2(1) + 1 = 3$$

This is why spin-1 is a **3-dimensional irrep**.

5. Example: Spin-$\frac{3}{2}$

This is the part confusing you.

Take:

$$S = \frac{3}{2}$$

Allowed m_S values:

$$-\frac{3}{2}, -\frac{1}{2}, +\frac{1}{2}, +\frac{3}{2}$$

So there are **4 states**.

$$2S + 1 = 2\left(\frac{3}{2}\right) + 1 = 4$$

The basis states are:

$$\left|\frac{3}{2}, +\frac{3}{2}\right\rangle$$

$$\left|\frac{3}{2}, +\frac{1}{2}\right\rangle$$

$$\left|\frac{3}{2}, -\frac{1}{2}\right\rangle$$

$$\left|\frac{3}{2}, -\frac{3}{2}\right\rangle$$

These four states together form the spin-$\frac{3}{2}$ irrep.

PART 4: Raising and Lowering Operators — The Practical Tools

You do not work with $\hat{S}_x$ and $\hat{S}_y$ directly. You use combinations:

$$\hat{S}_+ = \hat{S}_x + i\hat{S}_y \quad \text{(raising operator)}$$

$$\hat{S}_- = \hat{S}_x - i\hat{S}_y \quad \text{(lowering operator)}$$

What does S_+ do?

It raises the spin.

So:

$$S_+|\downarrow\rangle = |\uparrow\rangle$$

Meaning:

> if you are in the down state, raising operator moves you to up state.

But:

$$S_+|\uparrow\rangle = 0$$

because you are already at the top.

No higher state exists.

What does S_- do?

It lowers the spin.

$$S_-|\uparrow\rangle = |\downarrow\rangle$$

But:

$$S_-|\downarrow\rangle = 0$$

because you are already at the bottom.

Spin-1 example

Spin-1 has three ladder steps:

$$|1, +1\rangle$$

$$|1, 0\rangle$$

$$|1, -1\rangle$$

Now:

$$S_{-}|1, +1\rangle \rightarrow |1, 0\rangle$$

Again:

$$S_{-}|1, 0\rangle \rightarrow |1, -1\rangle$$

Again:

$$S_{-}|1, -1\rangle = 0$$

Bottom reached.

The BIG idea

All states in one irrep are connected by these ladder operators.

So if you know one state, you can generate all others.

Example:

Start from highest state:

$$|1, +1\rangle$$

Apply lowering repeatedly:

$$|1, +1\rangle \rightarrow |1, 0\rangle \rightarrow |1, -1\rangle$$

Entire irrep generated.

PART 5: Clebsch-Gordan Algebra — Combining Two Multiplets

The question: I have two particles, one with spin S1 and one with spin S2. What are the possible total spins of the combined system?

1. One particle vs two particles

A single spin-$\frac{1}{2}$ particle has:

$$|\uparrow\rangle, |\downarrow\rangle$$

That is one irrep.

Now suppose you have:

- particle 1 → spin-$\frac{1}{2}$
- particle 2 → spin-$\frac{1}{2}$

Question:

> What are the possible total spins when they combine?

This is called adding angular momentum.

2. Combine the states first

Each particle has 2 states.

So together:

$$2 \times 2 = 4$$

total states.

The four states are:

$$|\uparrow\uparrow\rangle$$

$$|\uparrow\downarrow\rangle$$

$$|\downarrow\uparrow\rangle$$

$$|\downarrow\downarrow\rangle$$

Now the question is:

> Can these 4 states be reorganized into irreps of SU(2)?

Answer: yes.

3. Result: they split into two groups

The 4 states separate into:

- one spin-0 irrep
- one spin-1 irrep

Written as:

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$$

Meaning:

> combining two spin-1/2 systems produces:
>
> - **"one singlet (spin 0)"**
> - **"one triplet (spin 1)"**

4. Why "singlet" and "triplet"?

Spin-0 has dimension:

$$2(0) + 1 = 1$$

So it has 1 state → singlet.

Spin-1 has dimension:

$$2(1) + 1 = 3$$

So it has 3 states → triplet.

Total:

$$1 + 3 = 4$$

Matches original 4 states.

5. The actual states

Triplet (spin-1):

$$|1,+1\rangle = |\uparrow\uparrow\rangle$$

$$|1,0\rangle = \frac{|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle}{\sqrt{2}}$$

$$|1,-1\rangle = |\downarrow\downarrow\rangle$$

These three states stay connected under rotations → one irrep.

Singlet (spin-0):

$$|0,0\rangle = \frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}}$$

This state forms its own separate irrep.

6. The big rule

When combining spins:

$$S_1 \otimes S_2$$

possible total spins are:

$$|S_1 - S_2|, |S_1 - S_2| + 1, \ldots, S_1 + S_2$$

7. Example: spin-1 with spin-1/2

Spin-1 has 3 states.

Spin-1/2 has 2 states.

Together:

$$3 \times 2 = 6$$

possible combined states.

These reorganize into:

- spin-1/2 irrep (2 states)
- spin-3/2 irrep (4 states)

So:

$$1 \otimes \frac{1}{2} = \frac{1}{2} \oplus \frac{3}{2}$$

Dimensions check:

$$2 + 4 = 6$$

Correct.

8. The intuition

Combining two spins is like combining two rotating objects.

The total rotation behavior can be broken into simpler irreducible pieces.

That decomposition is the Clebsch–Gordan decomposition.

Now:

$$E \otimes E$$

means:

> combine two copies of the 2D irrep.

Step 1: dimensions

E has dimension 2.

So:

$$2 \times 2 = 4$$

combined space is 4D.

Step 2: decomposition

The 4D space splits into:

$$A_1 \oplus A_2 \oplus E$$

Dimensions:

- A_1 → 1
- A_2 → 1
- E → 2

Total:

$$1 + 1 + 2 = 4$$

Correct.

The CG Coefficients Are The Expansion Coefficients

The combined states are linear combinations of product states. The CG coefficients are those coefficients. Written formally:

$$|S, M\rangle = \sum_{m_1, m_2} \langle S_1, m_1; S_2, m_2 | S, M\rangle \, |S_1, m_1\rangle |S_2, m_2\rangle$$

The number $\langle S_1, m_1; S_2, m_2 | S, M\rangle$ is the CG coefficient. It answers: how much of the product state $|m_1\rangle|m_2\rangle$ is in the combined state $|S, M\rangle$?

You have two spin-$\frac{1}{2}$ particles.

Each can be:

$$|\uparrow\rangle \quad \text{or} \quad |\downarrow\rangle$$

So together there are 4 product states:

$$|\uparrow\uparrow\rangle, \quad |\uparrow\downarrow\rangle, \quad |\downarrow\uparrow\rangle, \quad |\downarrow\downarrow\rangle$$

Now we want to reorganize these into states of definite TOTAL spin.

Step 1: Find the highest-spin state

The largest possible total M is:

$$+\frac{1}{2} + \frac{1}{2} = +1$$

Only one state has this:

$$|\uparrow\uparrow\rangle$$

So this must be:

$$|1, +1\rangle = |\uparrow\uparrow\rangle$$

This starts the spin-1 triplet.

Step 2: Lower it

Apply the lowering operator S_-.

It flips one spin down:

$$S_-(|\uparrow\uparrow\rangle) = |\downarrow\uparrow\rangle + |\uparrow\downarrow\rangle$$

This gives the middle triplet state:

$$|1,0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$$

The factor $1/\sqrt{2}$ is just normalization.

Step 3: Lowest triplet state

Lower again:

$$|1,-1\rangle = |\downarrow\downarrow\rangle$$

So the triplet is:

$$|1,+1\rangle = |\uparrow\uparrow\rangle$$

$$|1,0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$$

$$|1,-1\rangle = |\downarrow\downarrow\rangle$$

These 3 states form the spin-1 irrep.

Step 4: One state is still left

We used the symmetric combination:

$$|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle$$

But another independent combination exists:

$$|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle$$

This becomes:

$$|0,0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

This is the singlet (spin-0).

Final answer

The 4 original states split into:

Triplet (spin-1)

3 states:

$$|1,+1\rangle, \quad |1,0\rangle, \quad |1,-1\rangle$$

Singlet (spin-0)

1 state:

$$|0,0\rangle$$

What are CG coefficients here?

They are simply the numbers:

$$\frac{1}{\sqrt{2}}, \quad -\frac{1}{\sqrt{2}}$$

that tell you how to build total-spin states from product states.

Example:

$$|1,0\rangle = \frac{1}{\sqrt{2}}|\uparrow\downarrow\rangle + \frac{1}{\sqrt{2}}|\downarrow\uparrow\rangle$$

So the CG coefficients are:

$$\frac{1}{\sqrt{2}}, \quad \frac{1}{\sqrt{2}}$$

They are just expansion coefficients.

PART 6: Irreducible Operator Sets — Operators That Carry Symmetry Labels

Suppose you have orbitals transforming like the two components of an E irrep:

$$e_x,\ e_y$$

These are symmetry-related orbitals.

Then:

- $1e_x$ = first orbital of x-type
- $2e_x$ = second orbital of x-type

etc.

5. Meaning of each operator

T_{xx}

$$a^\dagger_{2e_x} a_{1e_x}$$

move electron:

$$1e_x \to 2e_x$$

T_{xy}

$$a^\dagger_{2e_x} a_{1e_y}$$

move electron:

$$1e_y \to 2e_x$$

T_{yx}

$$a^\dagger_{2e_y} a_{1e_x}$$

move electron:

$$1e_x \to 2e_y$$

T_{yy}

$$a^\dagger_{2e_y} a_{1e_y}$$

move electron:

$$1e_y \to 2e_y$$

6. Why grouped together?

Because symmetry rotations mix:

$$e_x \leftrightarrow e_y$$

So if symmetry transforms x-type into y-type, then:

$$T_{xx},\ T_{xy},\ T_{yx},\ T_{yy}$$

also transform into each other.

Thus they form one symmetry-connected operator family (IROP).

1. States can transform under symmetry

You already know:

$$|1,+1\rangle,\ |1,0\rangle,\ |1,-1\rangle$$

form a spin-1 multiplet.

Under rotations:

- they mix into each other
- ladder operators connect them

2. Operators can also behave the same way

The operators:

$$S^+,\ S^z,\ S^-$$

also transform into each other.

So they themselves form a spin-1 object.

This collection is called an:

Irreducible Operator Set (IROP)

Meaning:

> a set of operators that transforms together like one irrep.

3. Why?

Because commutators behave like ladder operations.

Example:

$$[S^z, S^+] = +S^+$$

$$[S^z, S^-] = -S^-$$

$$[S^+, S^-] \propto S^z$$

So:

- S^+
- S^z
- S^-

all connect to each other exactly like spin-1 states do.

Thus they form a spin-1 operator multiplet.

4. Main intuition

States:

$$|1, +1\rangle,\ |1, 0\rangle,\ |1, -1\rangle$$

are connected by ladder operators.

Similarly operators:

$$S^+,\ S^z,\ S^-$$

are connected by commutators.

Same structure.

5. Why this matters

Suppose you keep only:

$$S^+$$

but throw away:

$$S^z,\ S^-$$

Then symmetry is broken.

Because the full multiplet is incomplete.

6. Your VQE example

You had operators like:

$$T_{xx},\ T_{xy},\ T_{yx},\ T_{yy}$$

Symmetry generators can transform one into another:

$$T_{xx} \to T_{xy} \to T_{yy}$$

So these operators belong together as one symmetry multiplet (IROP).

7. Key physical message

If symmetry connects operators together, then:

> you should include the WHOLE connected set,
>
> not only some operators.

Otherwise the symmetry structure becomes incomplete.

That is the criticism of SymUCCSD mentioned there:

It keeps some symmetry-related operators but discards others.

So the full IROP structure is broken.

8. Super short summary

An IROP is:

> a group of operators that transform together under symmetry,
>
> exactly like states inside an irrep.

Example:

$$\{S^+, S^z, S^-\}$$

forms a spin-1 IROP.

The operators are connected by commutators just like states are connected by ladder operators.

The whole idea is:

Start with ONE operator.

Use symmetry generators to generate all symmetry-related operators.

The full connected family is the IROP.

This is exactly like generating all spin states from one highest-weight state.

BIG ANALOGY

For states

Start from:

$$|1, +1\rangle$$

Apply lowering operator:

$$S_-$$

Generate:

$$|1, 0\rangle$$

Apply again:

$$|1, -1\rangle$$

Now the multiplet is complete.

For operators

Start from one operator:

$$T_{xx}$$

Apply symmetry commutators.

Generate:

$$T_{xy},\ T_{yy},\ \dots$$

Eventually you get the full operator multiplet (IROP).

THE ALGORITHM

Step 1 — Pick a seed operator

Example:

$$T_{xx} = a^{\dagger}_{2e_x} a_{1e_x}$$

Meaning:

move electron:

$$1e_x \to 2e_x$$

This is your starting operator.

Step 2 — Find its symmetry label

Check how it behaves under symmetry.

This is done using commutators:

$$[S_z, F] = q_z F$$

If the result is proportional to itself:

$$[S_z, F] = (+1) F$$

then its label is:

$$q_z = +1$$

Exactly like states.

Step 3 — Apply ladder generators

Now apply symmetry generators repeatedly.

Like spin lowering:

$$S_-|1,+1\rangle \to |1,0\rangle$$

For operators:

$$[G, T_{xx}] \to T_{xy}$$

Then again:

$$[G, T_{xy}] \to T_{yy}$$

So symmetry rotates the operators into one another.

Step 4 — Continue until nothing new appears

Eventually:

$$[G, T] = 0$$

meaning:

you reached the top or bottom.

Then the operator multiplet is complete.

SIMPLE VISUAL PICTURE

Like spin states:

$$|1,+1\rangle \leftrightarrow |1,0\rangle \leftrightarrow |1,-1\rangle$$

Operator version:

$$T_{xx} \leftrightarrow T_{xy} \leftrightarrow T_{yy}$$

Symmetry connects them.

WHY IMPORTANT?

Suppose your ansatz keeps only:

$$T_{xx},\ T_{yy}$$

but removes:

$$T_{xy},\ T_{yx}$$

Then the symmetry ladder is broken.

The operators no longer form a complete symmetry multiplet.

This is the key criticism.

VQE INTERPRETATION

In VQE pools:

- operators should appear in symmetry-complete groups
- not partially

Otherwise:

- symmetry structure becomes inconsistent
- variational space becomes biased
- some symmetry directions are artificially removed

The key point: **you cannot have T_{xx} in your operator pool without T_{xy} and T_{yx} also being there, if you want an IROP.** SymUCCSD breaks the IROP structure by keeping T_{xx} and T_{yy} but discarding T_{xy} and T_{yx}. The remaining operators no longer form a proper IROP.

How To Build An IROP From A Seed Operator

This is the practical procedure from Weichselbaum B.2.1. Given one operator that you know is part of an IROP, generate the rest using commutators.

Step 1: Apply $\hat{S}_z$ to find the z-label of your seed operator:

$$[\hat{S}_z, \hat{F}] = q_z \hat{F}$$

Whatever number comes out is the z-label of $\hat{F}$.

Step 2: Apply $\hat{S}_+$ repeatedly to climb to the highest weight:

$$\hat{F}^q_{q_z+1} = \frac{1}{a_+}[\hat{S}_+, \hat{F}^q_{q_z}]$$

where a_+ is the same normalization factor as for states: $a_+ = \sqrt{q(q+1) - q_z(q_z+1)}$. This is Equation A.12 in Weichselbaum.

Step 3: Apply $\hat{S}_-$ to generate the remaining components going downward.

Step 4: When $[\hat{S}_+, \hat{F}^q_{q_{max}}] = 0$, you are at the top. When $[\hat{S}_-, \hat{F}^q_{q_{min}}] = 0$, you are at the bottom. The IROP is complete.

Worked Example For Your VQE Problem

Take the seed operator $T_{xx} = \hat{a}^\dagger_{2ex}\hat{a}_{1ex}$.

Step 1: Find its z-label under C3v. The C3v generators acting on the E components tell you T_{xx} has z-label $(q_z = +1)$ in some labeling.

Step 2: Apply the C3v analog of S_- — which is the generator that lowers the E component label from x to y:

$$[G_{x\to y}, T_{xx}] \propto T_{xy}$$

$$[G_{x\to y}, T_{xy}] \propto T_{yy}$$

Step 3: The resulting set $\{T_{xx}, T_{xy}, T_{yx}, T_{yy}\}$ organized into the CG combinations A_1, A_2, E is the complete set of IROPs in this sector.

The key point: **you cannot have T_{xx} in your operator pool without T_{xy} and T_{yx} also being there, if you want an IROP.** SymUCCSD breaks the IROP structure by keeping T_{xx} and T_{yy} but discarding T_{xy} and T_{yx}. The remaining operators no longer form a proper IROP.

PART 7: The Wigner-Eckart Theorem

Core idea of Wigner–Eckart theorem

Suppose you want matrix elements like:

$$\langle \text{final state} | \text{operator} | \text{initial state} \rangle$$

Normally, you think every matrix element is independent.

Wigner–Eckart says:

> NO — symmetry forces them to be related.

So every matrix element splits into:

$$\text{matrix element} = (\text{physics part}) \times (\text{symmetry part})$$

The formula

$$\langle q, q_z | F^{q_1}_{q_{1_z}} | q_2, q_{2z} \rangle = \langle q \| F^{q_1} \| q_2 \rangle \times C$$

means:

Part 2: Reduced matrix element

$$\langle q \| F^{q_1} \| q_2 \rangle$$

This is the actual physics.

It contains:

- Hamiltonian information
- molecule information
- dynamics

Importantly:

> it has NO z-labels.

So instead of many numbers, symmetry compresses everything into one number.

SIMPLE SPIN-1 EXAMPLE

Spin-1 has three states:

$$|1,+1\rangle, \quad |1,0\rangle, \quad |1,-1\rangle$$

A general operator would have:

$$3 \times 3 = 9$$

matrix elements.

Without symmetry:

you might think all 9 are unrelated.

But Wigner–Eckart says:

all 9 come from ONE reduced matrix element

plus fixed CG coefficients.

So symmetry determines almost everything automatically.

Example intuition

You already know:

$$S^z|1,m\rangle = m|1,m\rangle$$

So:

$$\langle 1,+1|S^z|1,+1\rangle = +1$$

$$\langle 1,0|S^z|1,0\rangle = 0$$

$$\langle 1,-1|S^z|1,-1\rangle = -1$$

Wigner–Eckart says:

> these numbers are NOT random.
>
> They come from:
>
> - **"one reduced matrix element"**
> - **"symmetry CG coefficients"**

BIG ADVANTAGE

Instead of storing many independent parameters:

symmetry ties them together.

So:

- fewer variational parameters
- fewer independent tensor entries
- fewer operator coefficients

This is why symmetry methods are powerful.

Your E ⊗ E example

Without symmetry:

A 2×2 operator has:

$$4$$

independent matrix elements.

But:

$$E \otimes E = A_1 \oplus A_2 \oplus E$$

splits into 3 symmetry sectors.

So only:

$$3$$

reduced matrix elements are needed.

Symmetry reconstructs the remaining structure automatically.

Super short summary

Wigner–Eckart theorem says:

$$\boxed{\text{matrix element} = \text{reduced physics number} \times \text{universal symmetry coefficient}}$$

where:

- reduced matrix element = problem-specific physics
- CG coefficient = fixed symmetry structure

So symmetry massively reduces the number of independent quantities.

PART 8: Putting It All Together For Your VQE Problem

STEP 1 — Your original operators

You have four possible excitation operators:

$$T_{xx},\ T_{yy},\ T_{xy},\ T_{yx}$$

They mean:

- excite from $x \to x$
- excite from $y \to y$
- excite from $x \to y$
- excite from $y \to x$

They are all symmetry-related pieces of the same object.

STEP 2 — What SymUCCSD does

SymUCCSD keeps only:

$$\theta_1 T_{xx} + \theta_2 T_{yy}$$

So it uses ONLY:

- T_{xx}
- T_{yy}

and throws away:

- T_{xy}
- T_{yx}

STEP 3 — Why this is bad

The FULL symmetry structure is:

$$E \otimes E = A_1 \oplus A_2 \oplus E$$

Meaning:

the four operators naturally split into:

- one A_1 sector
- one A_2 sector
- one E sector

But SymUCCSD only spans:

$$A_1 + E$$

The entire:

$$A_2$$

sector is missing.

STEP 4 — What is the missing A_2 operator?

It is:

$$T^{A_2} = \frac{1}{\sqrt{2}}(T_{xy} - T_{yx})$$

This antisymmetric combination is completely absent.

So:

> the optimizer can NEVER move in this direction.

Even if the true solution needs it.

VERY IMPORTANT IDEA

This is NOT an optimizer problem.

It is:

> a missing symmetry direction problem.

The variational space itself is incomplete.

STEP 5 — Proper IROP parametrization

Instead of optimizing raw operators separately:

$$T_{xx}, T_{yy}, T_{xy}, T_{yx}$$

you organize them by irreps:

$$A_1,\ A_2,\ E$$

Then the ansatz becomes:

$$\theta_{A_1} T^{A_1} + \theta_{A_2} T^{A_2} + \theta_E T^E$$

Now:

- all symmetry sectors exist
- the ansatz is complete
- symmetry is respected properly

HUGE IDEA

Inside each symmetry sector:

the ratios are FIXED by CG coefficients.

You do NOT optimize every operator independently.

You optimize only:

$$\theta_{A_1}, \theta_{A_2}, \theta_E$$

These are the reduced matrix elements.

Exactly like Wigner–Eckart theorem.

WHY GRADIENT BECOMES ZERO

Now comes the deepest point.

Even after including A_2:

you observed:

$$\langle HF|[H, T^{A_2}]|HF\rangle = 0$$

Why?

Because your molecular orbitals are Abelian-adapted.

That basis accidentally kills the reduced matrix element.

KEY INTERPRETATION

The CG coefficient is NOT zero.

Symmetry ALLOWS the transition.

But:

$$\text{reduced matrix element} = 0$$

So the gradient vanishes.

This is exactly what Wigner–Eckart predicts.

PHYSICAL INTERPRETATION

The symmetry channel exists.

But your orbital basis hides it.

So the optimizer cannot “see” the A_2 direction.

STEP 6 — Why orbital rotation fixes it

When you rotate orbitals into proper non-Abelian C_{3v} orbitals:

the reduced matrix element becomes nonzero.

Then:

$$\langle HF|[H, T^{A_2}]|HF\rangle \neq 0$$

Now the optimizer can access that symmetry channel.

Gradient appears.

Optimization proceeds.

THE BIG PICTURE

Your whole argument is:

SymUCCSD failure has TWO layers

Layer 1 — incomplete IROP

Missing operators:

$$T_{xy}, T_{yx}$$

So symmetry sectors are missing.

Layer 2 — bad orbital basis

Even with full IROP:

the reduced matrix element vanishes in Abelian orbi

So gradients disappear.

FINAL CONCEPTUAL SUMMARY

Clebsch–Gordan structure

tells you:

> how operators combine into symmetry sectors.

IROPs

tell you:

> which operators must appear together.

Wigner–Eckart theorem

tells you:

> optimizer parameters are reduced matrix elements,
> while symmetry fixes all internal ratios automatically.

Orbital rotation

changes:

$$\text{reduced matrix elements}$$

not the CG structure.

The claim

Fixing Gap 1 (restoring the complete IROP pool) automatically tells you the correct orbital rotation needed to fix Gap 2 (the gradient plateau). The rotation is not a separate guess — it falls out of the same CG structure.

STEP 1 — Missing operator problem

You originally had:

$$T_{xx},\ T_{yy}$$

but symmetry says the full structure is:

$$E \otimes E = A_1 \oplus A_2 \oplus E$$

The missing piece is:

$$T^{A_2} = \frac{1}{\sqrt{2}}(T_{xy} - T_{yx})$$

Notice the coefficients:

$$+\frac{1}{\sqrt{2}}, \quad -\frac{1}{\sqrt{2}}$$

These are CG coefficients.

They are fixed by C_{3v} symmetry.

STEP 2 — Why gradient vanished

You found:

$$\langle HF|[H, T^{A_2}]|HF\rangle = 0$$

Meaning:

the A_2 symmetry direction exists mathematically,

but your orbitals are aligned wrongly.

So the Hamiltonian cannot “see” this direction.

STEP 3 — What orbital rotation should fix it?

This is the key insight.

The SAME A_2 symmetry structure already tells you the correct mixing pattern.

The CG structure:

$$\frac{1}{\sqrt{2}}(xy - yx)$$

is antisymmetric.

So the orbital mixing must also be antisymmetric.

That gives the matrix:

$$\Omega_{A_2} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

What does this matrix do?

It mixes:

$$\phi_x \leftrightarrow \phi_y$$

in opposite directions.

Meaning:

$$\phi_x \to \phi_x + \epsilon \phi_y$$

$$\phi_y \to \phi_y - \epsilon \phi_x$$

That is exactly a rotation.

VERY IMPORTANT IDEA

You did NOT guess this rotation.

The symmetry itself generated it.

That is the beauty.

STEP 4 — Exponentiating gives rotation

You then build:

$$U = e^{\epsilon \Omega_{A_2}}$$

which becomes:

$$U = \begin{pmatrix} \cos\epsilon & \sin\epsilon \\ -\sin\epsilon & \cos\epsilon \end{pmatrix}$$

This is the standard 2D rotation matrix.

PHYSICAL INTERPRETATION

Originally:

the orbitals were aligned in an Abelian basis.

That accidentally killed:

$$h_{xy}$$

and therefore killed the gradient.

The rotation introduces mixing:

$$h_{xy} \neq 0$$

Now:

$$\langle HF|[H, T^{A_2}]|HF\rangle \neq 0$$

The optimizer can finally move.

THE DEEP UNIFICATION

This is your core conceptual contribution:

The same CG object plays two roles

Role 1 — operator construction

It tells you:

$$T^{A_2} = \frac{1}{\sqrt{2}}(T_{xy} - T_{yx})$$

which operators belong together.

Role 2 — orbital rotation generator

It tells you:

$$\Omega_{A_2} = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

which orbitals must mix.